homework_3

Part 1: Build the Dataset

```
library(dplyr)
library(tidyverse)
#creating data
day = 1:365
day
pm25 = runif(min=0, max=50, n=365)
pm25

risk = NA


seasons <- case_when(
  day %in% 1:90 ~ "winter",
  day %in% 91:180 ~"spring",
  day %in% 181:270 ~ "summer",
  day %in% 271:365 ~ "fall")
seasons

seasons <- as.factor(seasons)

data <- data.frame(seasons = seasons,
           pm25 = pm25,
           day = day,
           risk = risk)
data
```

Part 2 – Classify Risk with a For Loop

```r
# setting a named variable for the threshold makes the code easier to read
unhealthy = 35.4
moderate = 12
good = 0
max_unhealthy_streak = 0
current_streak = 0

for (i in 1:length(data$pm25)) {
  data$risk[i] <-
    ifelse(data$pm25[i] > unhealthy, "unhealthy",
           ifelse(data$pm25[i] >= moderate, "moderate", "good"))
  if (data$risk[i] == "unhealthy") {
    if (i == 1 || data$risk[i - 1] == "unhealthy") {
      current_streak <- current_streak + 1
    } else {
      current_streak <- 1
      }
  } else {
    current_streak <- 0
  }
  if (current_streak > max_unhealthy_streak)
      max_unhealthy_streak <- current_streak
  }

max_unhealthy_streak
```

```
[1] 4
```

Part 3 – Seasonal Summary with a For Loop

```r
# setting a named variable for the threshold makes the code easier to read
day_unhealthy <- c(winter = 0, spring = 0, summer = 0, fall = 0)


for (i in 1:length(data$day)) {
  if(data$risk[i] == "unhealthy") {
    seasons_i <- data$seasons[i]
```

```
    day_unhealthy[seasons_i] <- day_unhealthy[seasons_i] + 1
  }
}

day_unhealthy
```

```
winter spring summer   fall 
    26     24     21     24 
```

```
library(data.table)
```

```


Attaching package: 'data.table'


The following objects are masked from 'package:lubridate':

    hour, isoweek, mday, minute, month, quarter, second, wday, week,
    yday, year


The following object is masked from 'package:purrr':

    transpose


The following objects are masked from 'package:dplyr':

    between, first, last
```

```
seasonal_uh <- data.table(season = names(day_unhealthy),
                 unhealthy_days = as.integer(day_unhealthy))
seasonal_uh
```

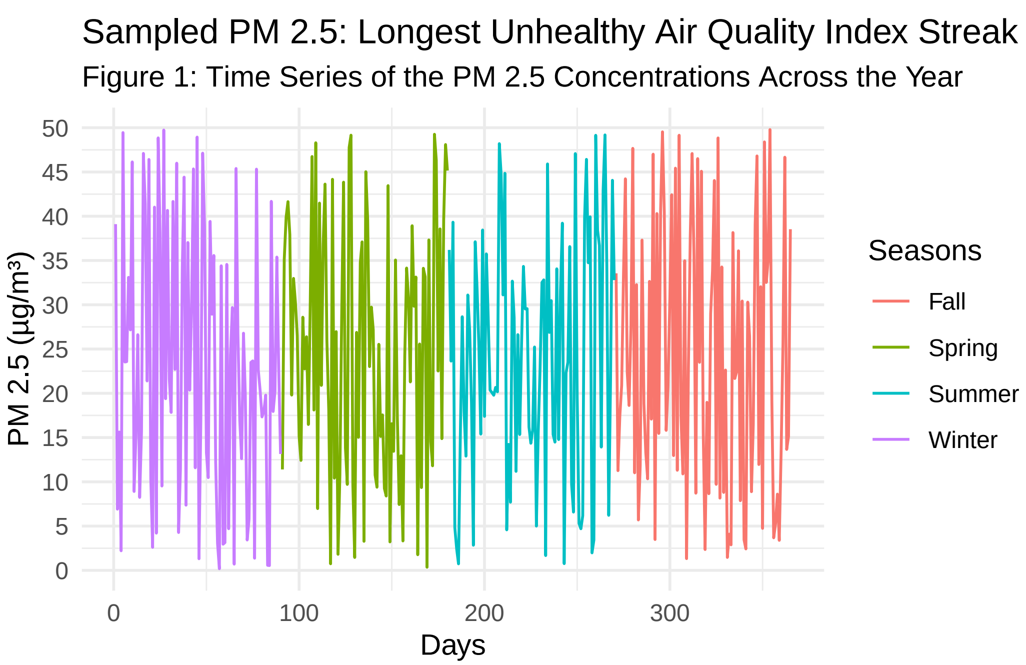
```
   season unhealthy_days
1: winter             26
2: spring             24
3: summer             21
4:   fall             24
```

Part 4 – Visualize and Report

```
library(ggplot2)

pm25_annual_season <- ggplot(data, aes(x=day, y=pm25, color=seasons)) +
  geom_line() +
  labs(colour = "Seasons",
       title = sprintf("Sampled PM 2.5: Longest Unhealthy Air Quality Index Streak was %d Day
       subtitle = "Figure 1: Time Series of the PM 2.5 Concentrations Across the Year",
       x = "Days",
       y = "PM 2.5 (µg/m³)") +
  scale_color_discrete(labels = c("Fall", "Spring", "Summer", "Winter")) +
  scale_y_continuous(breaks = seq(0, 50, by = 5)) +
  theme_minimal()

pm25_annual_season
```

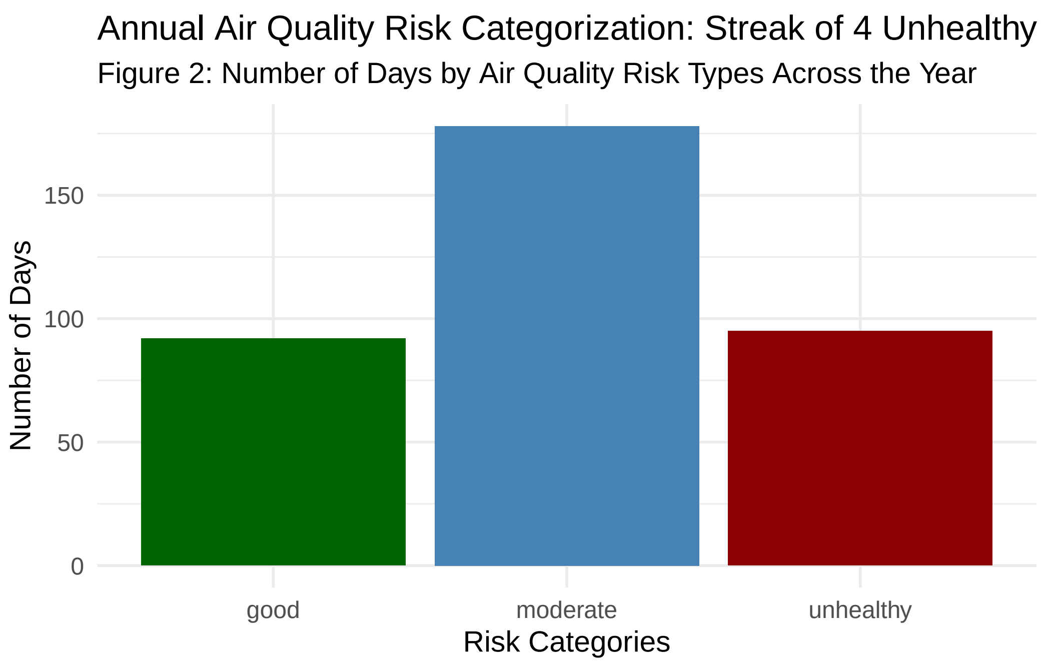



```
risk_counts <- data %>%
  group_by(risk) %>%
  summarise(n_days = n())
risk_counts
```

```
# A tibble: 3 x 2
```

```
  risk      n_days
  <chr>      <int>
1 good          92
2 moderate     178
3 unhealthy     95
```

```
risk_cat_plot <- ggplot(risk_counts, aes(x=risk, y=n_days, fill = risk)) +
  geom_col(fill = c("darkgreen", "steelblue", "darkred")) +
  labs(title = sprintf("Annual Air Quality Risk Categorization: Streak of %d Unhealthy Days"
       subtitle = "Figure 2: Number of Days by Air Quality Risk Types Across the Year",
       x = "Risk Categories",
       y = "Number of Days") +
 guides(fill="none") +
  theme_minimal()
risk_cat_plot
```



Results:

The seasonal 'for loop' analysis (`seasonal_uh`) revealed that spring had the highest number of unhealthy air quality index (AQI) days based on PM 2.5 concentrations (>35.4 µg/m³). This pattern is also evident in the annual PM 2.5 time series (Fig. 1), where spring concentrations fluctuate more dramatically than in other seasons, though the seasonal differences are difficult to distinguish visually from the time series alone. Winter had the second highest number of days (`seasonal_uh`). This seasonal pattern suggests that pollution sources are tied to late-winter transitioning into spring conditions. Potential sources include agricultural burning, wind-driven dust, or temperature-inversion events, while winter contributors likely include residential heating and stagnant air masses. From a management standpoint, these findings demonstrate the need for season-specific interventions rather than uniform year-round measures. Intervention strategies could involve enhanced public advisories and sensitive-group protections during spring and winter, stricter enforcement of emissions controls during high-risk months, and investment in real-time AQI alert systems for schools, elder-care facilities, and outdoor workers. Across the year, the number of moderate AQI days exceeded both the number of unhealthy and good days, accounting for nearly 50% of days, while unhealthy days accounted for approximately 30% (Fig. 2). The relatively small share of "good" days (<20%, Fig. 2) also indicates that this community is operating with little air-quality buffer, meaning that even modest increases in emissions or adverse weather could push a substantial number of days into unhealthy territory.

Put your quarto and a rendered version (pdf or html) in your GitHub repository for this assignment. You can use the same repository you created for HW2, but make you name the files clearly (or put in a new directory) so that I can find them. For example, you could put the quarto file in a new directory called "HW3" and name it something like "HW3_air_quality.qmd".

Turn in your GitHub repo link on Canvas